

Doc 5: Matrix Multiplication

Module 1: Linear algebra | Part 1 of 3: Learn (theory)

1. When can two matrices be multiplied?

Matrix A ($m \times n$) times matrix B ($n \times p$) works only when the **inner sizes match**. The result C has the **outer sizes**: $m \times p$.

$$(m \times n) \text{ times } (n \times p) = (m \times p)$$

(2×3) times (3×4) inner 3 and 3 match result 2×4; cost $2 \times 3 \times 4 = 24$ multiplications

(2×3) times (2×3) inner 3 and 2 differ not allowed

(1×3) times (3×1) inner 3 and 3 match result 1×1: a dot product (Doc 4)

(3×1) times (1×3) inner 1 and 1 match result 3×3 (an outer product, see page 2)

Why the rule? Each result cell pairs one row of A with one column of B. Their lengths must be equal, or the dot product does not exist. **Cost:** $(m \times n)(n \times p)$ takes $m \times n \times p$ multiplications, which is why big models need GPUs.

2. The grid method: how to compute the product

Put **A on the left**, **B on top**, and let the result C grow where they meet. Each cell of C is the **dot product** of the row of A to its left and the column of B above it.

		5	6
		7	8
1	2	19	22
3	4	43	50

Orange: row 1 of A and column 2 of B meet at the darker cell.

$$C_{12} = 1 \times 6 + 2 \times 8 = 6 + 16 = 22$$

Do this for every cell of C. A 2×2 result needs 4 dot products of length 2.

Habit: write the shape check first: $(2 \times 2)(2 \times 2)$ gives 2×2 .

3. Rules to remember

Order matters	AB and BA are usually different	BA may not even exist
Grouping is free	$(AB)C = A(BC)$	chains of layers can be regrouped
Spreads over sums	$A(B + C) = AB + AC$	like ordinary algebra
Transpose of a product	$(AB)^T = B^T A^T$	the order reverses
No cancelling	$AB = 0$ does not force $A = 0$ or $B = 0$	unlike ordinary numbers

Doc 5: Matrix Multiplication

Part 1 of 3: Learn (theory), continued

4. Four ways to see the same product

Using $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$, where $AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$. All four views give the same answer.

Dot view	each cell = (row of A) · (column of B)	$C_{12} = 1 \times 6 + 2 \times 8 = 22$
Row view	each row of C = a mix of the rows of B, weighted by a row of A	row 1 = $1 \times (5, 6) + 2 \times (7, 8) = (5, 6) + (14, 16) = (19, 22)$
Column view	each column of C = a mix of the columns of A, weighted by a column of B	column 1 = $5 \times (1, 3) + 7 \times (2, 4) = (5, 15) + (14, 28) = (19, 43)$
Outer view	AB = sum over k of (column k of A) times (row k of B) two 2×2 grids added, shown below	

$$\begin{array}{|c|c|} \hline 5 & 6 \\ \hline 15 & 18 \\ \hline \end{array} + \begin{array}{|c|c|} \hline 14 & 16 \\ \hline 28 & 32 \\ \hline \end{array} = \begin{array}{|c|c|} \hline 19 & 22 \\ \hline 43 & 50 \\ \hline \end{array}$$

col 1 of A ×
row 1 of B
col 2 of A ×
row 2 of B
AB

Best uses: the dot view for one cell, the column view to understand what a matrix does to a vector ($A \times$ is a mix of the columns of A).

5. Special matrices worth knowing

Identity I	$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$	$AI = IA = A$: changes nothing
Diagonal D	$\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$	D A scales the rows of A; A D scales its columns
Swap P	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	P A swaps the rows of A; A P swaps its columns
Zero matrix	all entries 0	A times the zero matrix is the zero matrix

6. Where machine learning uses it

- **One layer:** $y = Wx + b$. Each row of W is one neuron, so one product runs all neurons at once.
- **A batch:** XW processes many examples in one product (rows of X are examples).
- **Stacked layers:** $W_2(W_1x) = (W_2W_1)x$. Without a ReLU in between, two layers collapse into one (Example 8).
- **Embeddings:** a one-hot vector times an embedding matrix picks out one row: a table lookup (Example 9).
- **Attention** (Doc 29): the scores are QK^T , a table of dot products of every query with every key (Example 9).

Doc 5: Matrix Multiplication

Part 2 of 3: Worked examples. Every cell filled in. Read, then cover the answers and redo them.

Example 1: Sales revenue (a real-life product)

		10
		5
2	1	25
0	3	15
4	2	50

Rows of the left matrix are stores A, B, C. Its columns are shirts and hats sold.

The top column is the price list: shirt 10, hat 5.

Store A: $2 \times 10 + 1 \times 5 = \mathbf{25}$. Store B: $0 \times 10 + 3 \times 5 = \mathbf{15}$. Store C: $4 \times 10 + 2 \times 5 = \mathbf{50}$.

Shape: $(3 \times 2)(2 \times 1) = 3 \times 1$: one revenue per store.

Example 2: Two 2×2 matrices (with negatives)

		4	1
		2	5
1	-2	0	-9
3	0	12	3

$$C_{11} = 1 \times 4 + (-2) \times 2 = 4 - 4 = \mathbf{0}$$

$$C_{12} = 1 \times 1 + (-2) \times 5 = 1 - 10 = \mathbf{-9}$$

$$C_{21} = 3 \times 4 + 0 \times 2 = 12 + 0 = \mathbf{12}$$

$$C_{22} = 3 \times 1 + 0 \times 5 = 3 + 0 = \mathbf{3}$$

Example 3: A (2×3) times B (3×2) gives a 2×2

			3	1
			2	1
			1	0
1	0	2	5	1
-1	3	1	4	2

$$C_{11} = 1 \times 3 + 0 \times 2 + 2 \times 1 = 5$$

$$C_{12} = 1 \times 1 + 0 \times 1 + 2 \times 0 = 1$$

$$C_{21} = (-1) \times 3 + 3 \times 2 + 1 \times 1 = 4$$

$$C_{22} = (-1) \times 1 + 3 \times 1 + 1 \times 0 = 2$$

Shape check: $(2 \times 3)(3 \times 2) = 2 \times 2$.

Example 4: 3×3 times 3×3 (27 multiplications)

			1	2	0
			0	1	1
			3	0	1
1	0	2	7	2	2
0	1	-1	-3	1	0
2	1	0	2	5	1

9 cells, each a dot product of length 3.

Row 2 of A is (0, 1, -1); column 1 of B is (1, 0, 3):

$$C_{21} = 0 \times 1 + 1 \times 0 + (-1) \times 3 = \mathbf{-3}$$

Row 1 with column 3: $1 \times 0 + 0 \times 1 + 2 \times 1 = \mathbf{2}$

Tip: zeros make a cell quick. Skip those products.

Doc 5: Matrix Multiplication

Part 2 of 3: Worked examples, continued

Example 5: Row view and column view of the same product

			2	1
			0	3
			1	-1
1	2	0	2	7
3	-1	2	8	-2

Row view. Row 1 of A is (1, 2, 0), so row 1 of C mixes the rows of B:

$$1 \times (2, 1) + 2 \times (0, 3) + 0 \times (1, -1) = (2, 1) + (0, 6) + (0, 0) = (2, 7)$$

Column view. Column 1 of B is (2, 0, 1), so column 1 of C mixes the columns of A:

$$2 \times (1, 3) + 0 \times (2, -1) + 1 \times (0, 2) = (2, 6) + (0, 0) + (0, 2) = (2, 8)$$

Both match the grid: row 1 is (2, 7); column 1 is (2, 8).

Example 6: Special matrices in action

D M

			1	2
			3	4
2	0	2	4	
0	3	9	12	

M D

			2	0
			0	3
1	2	2	6	
3	4	6	12	

P M

			1	2
			3	4
0	1	3	4	
1	0	1	2	

M P

			0	1
			1	0
1	2	2	1	
3	4	4	3	

$M = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. **D M**: row 1 doubled, row 2 tripled. **M D**: column 1 doubled, column 2 tripled. **P M**: rows swapped. **M P**: columns swapped. The matrix on the **left** acts on rows; on the **right** acts on columns.

Example 7: A neural-network layer, $y = \text{ReLU}(W x + b)$

			1
			2
			3
2	-1	0	0
1	3	1	10

Neuron	1	2
$W x$ (from the grid)	0	10
Bias b	1	-12
$z = W x + b$	1	-2
$\text{ReLU}(z)$	1	0

Neuron 2 gets $z = -2$, and ReLU clips it to 0. Both neurons were computed by the single product $W x$.

Doc 5: Matrix Multiplication

Part 2 of 3: Worked examples, continued

Example 8: Two layers without ReLU collapse into one

Route 1, step 1: $W1 \times x$

		1
		3
1	2	7
0	1	3

Step 2: $W2 (W1 \times x)$

		7
		3
2	0	14
1	1	10

Route 2, step 1: $W2 W1$

		1	2
		0	1
2	0	2	4
1	1	1	3

Step 2: $(W2 W1) \times x$

		1
		3
2	4	14
1	3	10

$W1 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $W2 = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$, $x = (1, 3)$. Both routes give **(14, 10)**. So a stack of purely linear layers is just one matrix $W2 W1$. The ReLU between layers is what makes deep networks more powerful.

Example 9: Two ML uses of matrix products

Embedding lookup: one-hot $\times E$

				1	2	0
				3	1	4
				0	5	2
				2	2	1
0	0	1	0	0	5	2

Attention scores: Q times K -transpose

		1	2	0
		1	0	3
1	0	1	2	0
0	1	1	0	3
1	1	2	2	3

Left: the 1 sits in position 3 of the one-hot row, so the product is exactly row 3 of E : (0, 5, 2). A matrix product can act as a table lookup.

Right: cell (i, j) is the dot product of query i with key j . Row 3 = (2, 2, 3): query 3 matches key 3 best. This 3×3 table of scores is the heart of attention.

Doc 5: Matrix Multiplication

Part 3 of 3: Practice. Use a pencil; yellow cells are yours. Name: _____ Date: _____

A. Can they be multiplied? Result shape? How many multiplications?

Product	Allowed? (Yes/No)	Result shape	Multiplications
(3×2) times (2×4)			
(2×3) times (2×3)			
(1×4) times (4×1)			
(4×1) times (1×4)			
(2×5) times (5×3)			

B and C. Fill the result grid (blue = A, purple = B)

B. (2×2) times (2×2)

		1	4
		5	2
2	1		
0	3		

C. (2×3) times (3×2)

			1	2
			4	0
			-1	3
2	0	1		
1	-1	3		

D. (3×2) times (2×3): the result is 3×3

		1	0	2
		2	1	-1
1	2			
0	1			
3	-1			

Nine dot products, each of length 2.

Work one row of C at a time.

Doc 5: Matrix Multiplication

Practice, continued

E. 3×3 times 3×3 (27 multiplications)

			1	1	0
			2	0	-1
			0	4	1
2	0	1			
1	-1	0			
0	3	1			

Tip: skip products with a zero.

F. Matrix times vector = a mix of columns

Compute $A \times x$ with the grid. Then check it by mixing the columns of A: $A \times x = 3 \times (\text{column 1}) + 2 \times (\text{column 2})$.

		3
		2
2	1	
0	3	
1	-1	

Entry	1	2	3
3 × column 1 of A			
2 × column 2 of A			
Sum = $A \times x$			

G. Row view and column view

Fill the grid for $C = AB$. Then rebuild row 1 of C from the rows of B, and column 2 of C from the columns of A.

			1	2
			3	0
			0	1
1	0	2		
2	1	0		

Row view	1	2
1 × row 1 of B		
0 × row 2 of B		
2 × row 3 of B		
Row 1 of C		

Column view	1	2
2 × column 1 of A		
0 × column 2 of A		
1 × column 3 of A		
Column 2 of C		

Doc 5: Matrix Multiplication

Practice, continued

H. Special matrices: $M = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$, $D = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}$, $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

D M

		2	1
		0	3
1	0		
0	-2		

M D

		1	0
		0	-2
2	1		
0	3		

P M

		2	1
		0	3
0	1		
1	0		

M P

		0	1
		1	0
2	1		
0	3		

What does D do when it is on the left? _____ On the right? _____

I. The outer-product view: add two grids

1	0		
2	1		

×

3	1		
0	2		

A

B

+

=

(1, 2) times (3, 1)

(0, 1) times (0, 2)

AB

$AB = (\text{column 1 of } A)(\text{row 1 of } B) + (\text{column 2 of } A)(\text{row 2 of } B)$. Each grid holds every entry of the column times every entry of the row.

J. Two layers collapse: $W1 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$, $W2 = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$, $x = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

W1 x

		1
		2
2	1	
0	1	

W2 (W1 x): copy W1 x on top

1	0		
3	1		

W2 W1

		2	1
		0	1
1	0		
3	1		

(W2 W1) x

		1
		2

Do the two routes give the same final vector? _____ What is it? _____

Doc 5: Matrix Multiplication

Practice, continued

K. One-hot times an embedding matrix (rows of E are words 1 to 4)

				1	0	2
				3	1	0
				0	2	2
				1	1	4
0	1	0	0			

Which row of E does the product pick? Row _____

One-hot vector for word 4: (__, __, __, __)

L. Attention scores: Q times K-transpose

		1	2	0
		1	0	4
1	0			
0	1			
1	2			

Each cell is a dot product of a query row with a key column.

The highest score is in query ____, key ____.

M. Word problem: two price lists at once

Rows of Q are 2 stores; columns are shirts and hats sold. Columns of P are the regular and sale price lists (rows: shirt, hat). The product $R = Q P$ has one row per store and one column per price list.

		12	10
		5	4
3	2		
1	4		

Store 2 revenue at sale prices: _____

Which store earns more at regular prices? _____

Total revenue of both stores at regular prices: _____

Doc 5: Matrix Multiplication

Practice, continued

N. True or False?

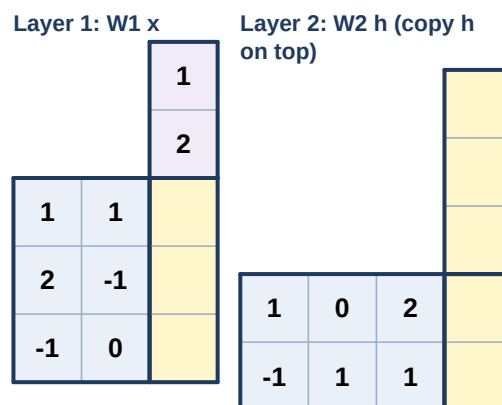
Statement	True / False
1. A is 2×3 and B is 3×2 . Both AB and BA are defined.	
2. $AB = BA$ for any two 2×2 matrices.	
3. $(AB)C = A(BC)$ whenever the shapes allow it.	
4. Multiplying a matrix by the identity I leaves it unchanged.	
5. If AB is the zero matrix, then A or B must be zero.	
6. A (4×3) times B (3×5) needs $4 \times 3 \times 5 = 60$ multiplications.	

O. Detective work

Question	Answer
1. A is 3×4 and AB is 3×5 . What shape is B?	
2. $A = \begin{bmatrix} 1 & x \\ 2 & 1 \end{bmatrix}$, $B = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ as a column, $AB = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$. Find x.	
3. A is 4×2 and B is 2×6 . How many multiplications does AB need?	

P. Capstone: a two-layer network, forward pass

$x = (1, 2)$. Layer 1: $W1 (3 \times 2) = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ -1 & 0 \end{bmatrix}$, $b1 = (0, 1, 0)$, then ReLU. Layer 2: $W2 (2 \times 3) = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & 1 \end{bmatrix}$, $b2 = (1, 2)$, no ReLU. Fill in every yellow cell.



Layer 1	1	2	3
b1	0	1	0
$z1 = W1 \times x + b1$			
$h = \text{ReLU}(z1)$			

Layer 2	1	2
b2	1	2
$y = W2 \times h + b2$		

Doc 5: Matrix Multiplication

Answer key (page 1 of 2): check your work only after finishing

A. Shape check

Product	Allowed?	Result shape	Multiplications
(3×2) times (2×4)	Yes	3×4	24
(2×3) times (2×3)	No	none	none
(1×4) times (4×1)	Yes	1×1	4
(4×1) times (1×4)	Yes	4×4	16
(2×5) times (5×3)	Yes	2×3	30

B, C, D and E

B

		1	4
		5	2
2	1	7	10
0	3	15	6

C

		1	2
		4	0
		-1	3
2	0	1	1
1	-1	3	-6

D

		1	0	2
		2	1	-1
1	2	5	2	0
0	1	2	1	-1
3	-1	1	-1	7

E

			1	1	0
			2	0	-1
			0	4	1
2	0	1	2	6	1
1	-1	0	-1	1	1
0	3	1	6	4	-2

F. Matrix times vector

		3
		2
2	1	8
0	3	6
1	-1	1

Entry	1	2	3
3 × column 1	6	0	3
2 × column 2	2	6	-2
Sum = A x	8	6	1

G. Row view and column view

			1	2
			3	0
			0	1
1	0	2	1	4
2	1	0	5	4

Row view	1	2
1 × row 1 of B	1	2
0 × row 2 of B	0	0
2 × row 3 of B	0	2
Row 1 of C	1	4
Column view	1	2
2 × col 1 of A	2	4
0 × col 2 of A	0	0
1 × col 3 of A	2	0
Column 2 of C	4	4

Row 1 of C = (1, 4) and column 2 of C = (4, 4): both views agree with the grid.

H. Special matrices

D M	M D	P M	M P																																																																
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D on the left scales the rows (row 2 times -2). D on the right scales the columns (column 2 times -2). P swaps rows on the left, columns on the right.

Doc 5: Matrix Multiplication

Answer key (page 2 of 2)

I. Outer products

$$\begin{array}{|c|c|} \hline 3 & 1 \\ \hline 6 & 2 \\ \hline \end{array} + \begin{array}{|c|c|} \hline 0 & 0 \\ \hline 0 & 2 \\ \hline \end{array} = \begin{array}{|c|c|} \hline 3 & 1 \\ \hline 6 & 4 \\ \hline \end{array}$$

$\begin{array}{|c|c|} \hline 3 & 1 \\ \hline 6 & 2 \\ \hline \end{array}$ col 1 $\times$ row 1 $\begin{array}{|c|c|} \hline 0 & 0 \\ \hline 0 & 2 \\ \hline \end{array}$ col 2 $\times$ row 2 $\begin{array}{|c|c|} \hline 3 & 1 \\ \hline 6 & 4 \\ \hline \end{array}$ AB

J. Two layers collapse

W1 x W2 (W1 x) W2 W1 (W2 W1) x

$$\begin{array}{|c|c|c|} \hline & & 1 \\ \hline & & 2 \\ \hline 2 & 1 & 4 \\ \hline 0 & 1 & 2 \\ \hline \end{array} \quad \begin{array}{|c|c|c|} \hline & & 4 \\ \hline & & 2 \\ \hline 1 & 0 & 4 \\ \hline 3 & 1 & 14 \\ \hline \end{array} \quad \begin{array}{|c|c|c|c|} \hline & & 2 & 1 \\ \hline & & 0 & 1 \\ \hline 1 & 0 & 2 & 1 \\ \hline 3 & 1 & 6 & 4 \\ \hline \end{array} \quad \begin{array}{|c|c|c|} \hline & & 1 \\ \hline & & 2 \\ \hline 2 & 1 & 4 \\ \hline 6 & 4 & 14 \\ \hline \end{array}$$

Both routes give (4, 14): yes, the same vector.

K, L and M

K L M

$$\begin{array}{|c|c|c|c|c|c|} \hline & & & & 1 & 0 & 2 \\ \hline & & & & 3 & 1 & 0 \\ \hline & & & & 0 & 2 & 2 \\ \hline & & & & 1 & 1 & 4 \\ \hline 0 & 1 & 0 & 0 & 3 & 1 & 0 \\ \hline \end{array} \quad \begin{array}{|c|c|c|c|c|c|} \hline & & & & 1 & 2 & 0 \\ \hline & & & & 1 & 0 & 4 \\ \hline 1 & 0 & 1 & 2 & 0 \\ \hline 0 & 1 & 1 & 0 & 4 \\ \hline 1 & 2 & 3 & 2 & 8 \\ \hline \end{array} \quad \begin{array}{|c|c|c|c|c|c|} \hline & & & & 12 & 10 \\ \hline & & & & 5 & 4 \\ \hline 3 & 2 & 46 & 38 \\ \hline 1 & 4 & 32 & 26 \\ \hline \end{array}$$

K: row 2 of E = (3, 1, 0); word 4 needs (0, 0, 0, 1). L: highest score 8, query 3 and key 3. M: store 2 at sale prices 26; store 1 earns more at regular prices (46 vs 32); total regular = 46 + 32 = 78.

N and O

N: 1 True (AB is 2×2 , BA is 3×3). 2 False (e.g. $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ differs from the reverse order). 3 True. 4 True. 5 False (e.g. $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ is the zero matrix). 6 True.

O: 1: B is 4×5 . 2: row 1 gives $2 + x = 5$, so $x = 3$ (row 2 checks: $4 + 1 = 5$). 3: $4 \times 2 \times 6 = 48$.

P. Capstone

W1 x W2 h

$$\begin{array}{|c|c|c|} \hline & & 1 \\ \hline & & 2 \\ \hline 1 & 1 & 3 \\ \hline 2 & -1 & 0 \\ \hline -1 & 0 & -1 \\ \hline \end{array} \quad \begin{array}{|c|c|c|c|} \hline & & & 3 \\ \hline & & & 1 \\ \hline & & & 0 \\ \hline 1 & 0 & 2 & 3 \\ \hline -1 & 1 & 1 & -2 \\ \hline \end{array}$$

$W1 \ x = (3, 0, -1)$
 $z1 = W1 \ x + b1 = (3, 1, -1)$
 $h = \text{ReLU}(z1) = (3, 1, 0)$
 $W2 \ h = (3, -2)$
 $y = W2 \ h + b2 = (4, 0)$

Key takeaways: inner sizes must match; the result has the outer sizes; each cell is a dot product; order matters; the left matrix acts on rows, the right on columns; stacked linear layers collapse without a ReLU; a layer is $W \ x + b$.

Next: Doc 6 - Transpose, inverse, determinant.